

Dang Hyeona

Backend developer turning operational needs into product features

I build and operate Java/Spring Boot servers, turning complex requirements into API, data, and shared product capabilities.

Java

Spring Boot

Backend

Operation

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Development flow through work experience

A path from research and education to production development focused on turning operational needs into product capabilities

2019

Summer intern · research trainee

ETRI Research Intern

Frame-editing UI and background-removal experiment flow for an AI-based video editing project

Tech

Python 3.6, Tkinter, OpenCV, GrabCut
Pillow, NumPy, Keras, scikit-learn

Role

Video editing tool development
2-person project team

Work

- Built a Tkinter-based frame editing UI
- Implemented GrabCut-based manual background removal
- Experimented with CNN-based automatic background removal
- Adjusted learning layers and input sizes
- Configured a flow where users can select experiment conditions

Outcome

- Compared manual editing and automatic detection results using the F-measure
- Connected the experiment to a tool flow where users can select conditions and review results

2021-22

Contract freelance

SSAFY Project Practice Coach

Student project coaching · coach operations standards

Tech

Project and presentation coaching, Git, Linux/CLI deployment
Server operation support, operation documentation

Role

Project practice coach for SSAFY 6th-7th cohorts
New-coach OJT and work guidance

Work

- Coached 66 teams from planning through presentation across six project cycles
- Supported coaching and operations for Samsung Electronics-linked projects
- Organized reporting templates, document workflows, and best-case samples for practice coaches
- Led about 15 live development sessions and formalized the workflow as an operation manual
- Operated open-source management and Twin Campus Seoul responsibilities

Outcome

- Defined project operation scope and new-coach work standards for SSAFY 6th-7th cohorts
- Organized external-linked projects, live sessions, and operation documentation into formal processes

2024-NOW

Full-time · senior

Bigxdata Data Solutions Team / Senior

Expanded an initial BI portal into a data portal product while building product capabilities and operational foundations.

Tech

Java 8-17, Spring Boot 2.x-3.4.5, JPA/MyBatis/QueryDSL
PostgreSQL/Oracle/MariaDB/MySQL, Swagger

Role

Product features, shared management, and data catalog ownership
Product structure transition and diverse operating environments

Work

- Built a DBMS management feature end to end, from planning and design through development and testing.
- Supported PostgreSQL and Oracle with Flyway-based database change management.
- Built shared email delivery and template features while reviewing impact across diverse operating environments.
- Covered patch-release design, task assignment, and implementation for about three to four months.
- Implemented visualization and DBMS features during the product structure transition, then expanded ownership into shared management, data catalog, and curation features.

Outcome

- Connected DBMS management capabilities to product features usable across diverse operating environments.
- Contributed to a product structure transition that improved feature extensibility.
- Built a foundation for data portal expansion through shared management and catalog capabilities.

Technical Skills

Technologies organized by usage level across work and project experience.

USAGE LEVEL

- 1 bar
Basic build
- 2 bars
Feature build
- 3 bars
Project
- 4 bars
Work use
- 5 bars
Improve/Ops

LANGUAGE

Java



Built Java 8-17 server logic and verified runtime errors and response model impact.

Java 8-17

Server Logic

Debugging

FRAMEWORK

Spring Boot



Developed Spring Boot 2-3.4.5 product features with API, data access, and WAS structure in mind.

Boot 2-3.4

JPA/MyBatis

Tomcat

API DESIGN

REST API



Handled API design through implementation, including auth and environment-based API flows.

Auth Branch

Condition API

Swagger

DATABASE

DBMS



Worked on service DB design, migration, and query improvement with PostgreSQL, MariaDB, and MySQL.

PostgreSQL

Migration

Index

WEB SERVER

Nginx



Reviewed server version upgrades and security header settings for service operations.

Version Response

Security Headers

Ops Check

CLOUD

AWS



Worked with load balancing, RDS, and monitoring setup for development and operations environments.

Load Balancing

RDS

Monitoring

VERSION CONTROL

Git



Created and applied team conventions while adapting to established production conventions.

Convention

Commit Scope

Push

COLLABORATION TOOL

Atlassian



Structured Jira and Confluence information and organized recurring team-level shared updates.

Jira

Confluence

Team Share

DEVELOPMENT SUPPORT

AI-assisted Development



Organized shared AI setup and skill structure that became an actual team convention.

Codex

Claude

Documentation

PRODUCT SCALING EXPERIENCE

PRODUCT

Productization Experience

Participated in expanding an existing project into a productized portal service by organizing feature flows.

Productized Service

Multiple Environments

Productization

ARCH

Scalable Structure Transition

Developed an extensible structure by considering feature boundaries and shared flows during a product structure transition.

Product Scaling

Structure Transition

Shared Area

COMMON

Shared & Hub Feature Contribution

Expanded my ownership from specific feature modules into shared areas and hub-like features.

Shared Feature

Hub-like Feature

Extension

SYSTEM

Team Development System

Organized AI usage flows as a team convention and continuously managed Confluence knowledge and onboarding materials.

AI Convention

Confluence

Onboarding Materials

Safers

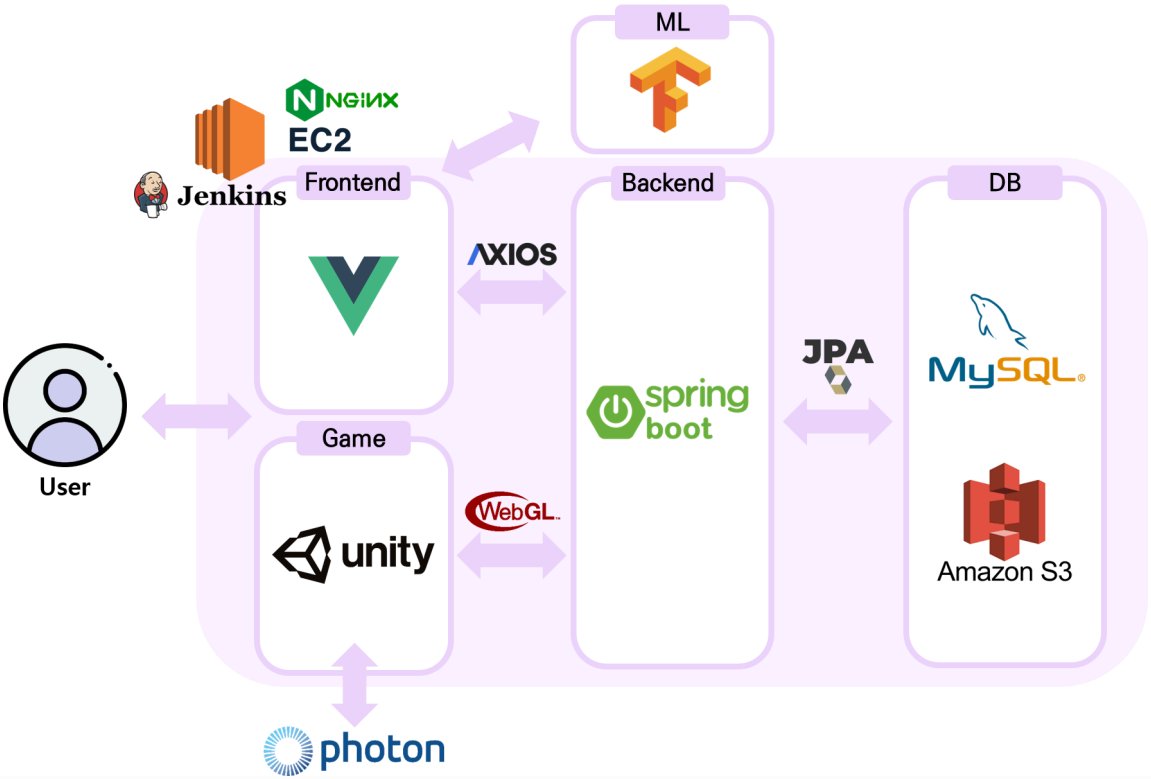
A service that delivers environmental information through interactive experiences.

Period
Oct 12–Nov 19, 2021 (6 weeks)

Domain
Web · Metaverse

Team
Five-person team · Team lead

Role
Backend · Infrastructure · ML · Presentation



Technologies and versions

The main technologies and runtime environment used for the project.

Language	Java 8
Backend	Spring Boot 2.5.6 · JPA
Web · platform	Vue 2.6.14 · Unity 2019.4.32f1 · Photon 2.38
Data · infra	MySQL 8.0.22 · AWS EC2/S3 · Jenkins 2.303.2 · nginx 1.18.0

Key responsibilities

Backend · API · Deployment · ML

- Designed data structures for users, animals, maps, missions, and logs
- Contributed to the design and implementation of core APIs for users, boards, and Unity integration
- Connected Unity WebGL with server data
- Configured the service environment on AWS EC2 and S3
- Built the Jenkins and nginx CI/CD flow
- Applied ML-based image mission validation

Safers

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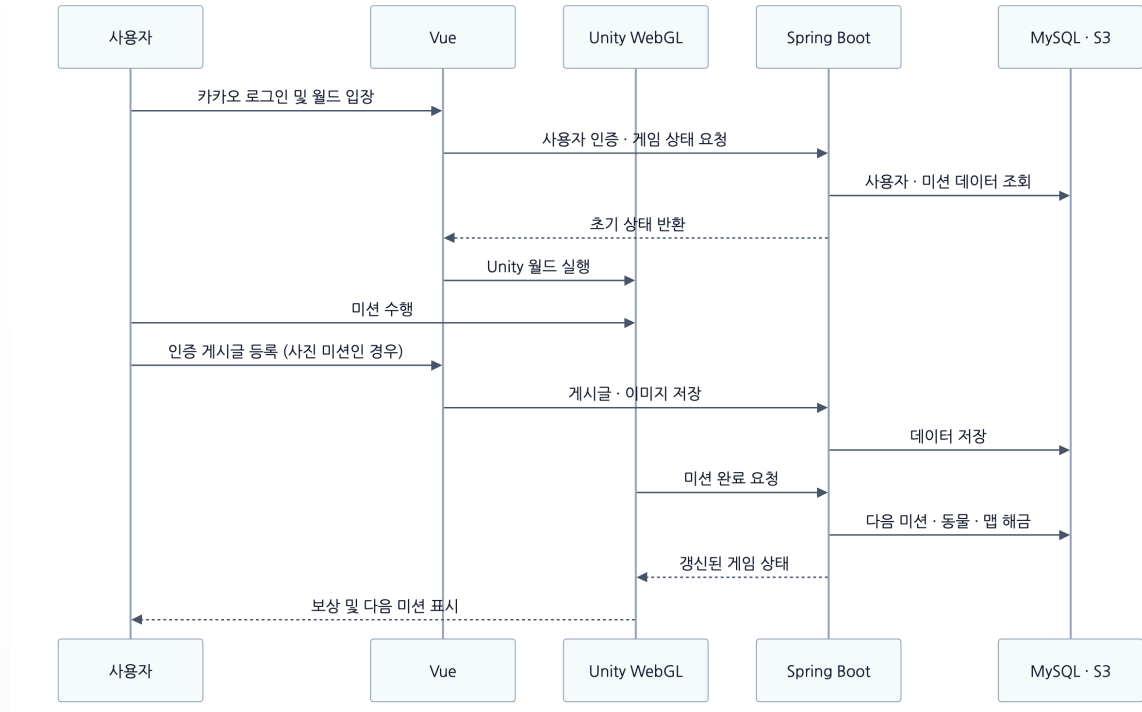
Team

Five-person team · Team lead

Role

Backend · Infrastructure · ML · Presentation

API design across the service flow



01

Shared code and content data design

Designed a structure that supports future service expansion.

- Designed four shared-code groups and fourteen detail codes
- Defined data and initial values for three maps, thirty-two animals, and thirty missions
- Connected state values and next-mission relations to the API flow

02

Unity WebGL and server integration design

- Connected the user-authentication and initial-state flow after login
- Integrated user, mission, and animal data between Vue, Unity WebGL, and Spring Boot APIs
- Structured the API flow from mission-result storage to next-content state updates

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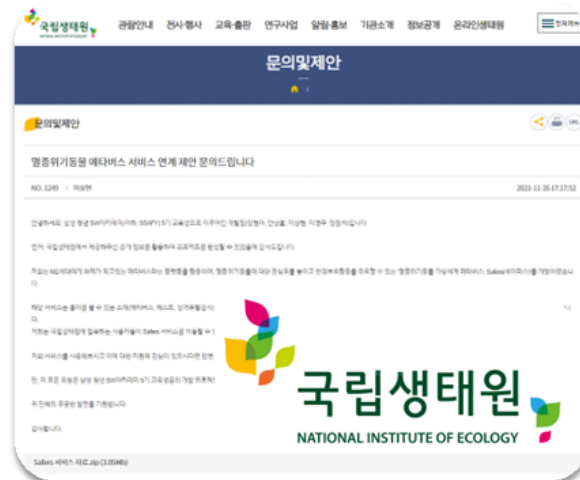
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Role

Backend • Infrastructure • ML • Presentation

Outcome and external validation



01

Operational foundation

- Resolved Jenkins-SSH issues and established an automated deployment flow
- Documented the reproduction, fix, and redeployment flow for deployment-version errors
- Documented deployment response procedures for repeated team use

02

External validation and project result

- Confirmed a positive response from the National Institute of Ecology
- Communicated externally about using IUCN Red List information
- Placed first among eight teams in the SSAFY 5th autonomous project
- Won nationally as one of the top eight teams across four campuses and 750 participants

Node-RED Motion Pose

A public project that made camera-based motion recognition usable in Node-RED and SmartThings environments.

Period

Aug 23–Oct 8, 2021 (7 weeks)

Domain

AI · IoT · Node-RED

Team

5-person team · Team lead

Role

Node-RED development · Presentation · Distribution

Technologies and versions

The technologies used for motion recognition and Node-RED distribution.

Runtime	Node.js 14.17.3 · Node-RED 2.0.6
Recognition	MediaPipe BlazePose · Hands
Communication · server	Socket.IO 4.2.0 · Express 4.17.1
Distribution	npm package 1.1.3 · Node-RED Library

Key responsibilities

Hand recognition · Node-RED · Distribution · Contribution

- Built the hand-recognition flow for the Hands Pose Node
- Implemented hand registration and result discovery
- Packaged the features as reusable Node-RED nodes
- Published nine nodes to npm and the Node-RED Library
- Contributed PRs #46 and #47 to Samsung Automation Studio

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A public project that made camera-based motion recognition usable in Node-RED and SmartThings environments.

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Domain

AI · IoT · Node-RED

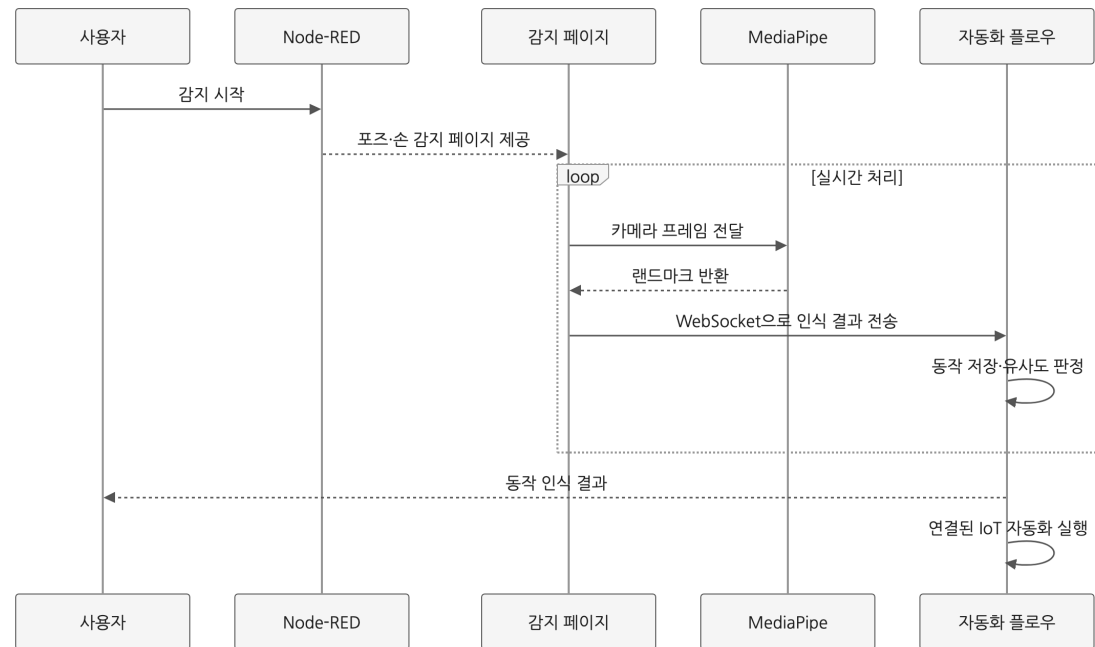
Team

5-person team · Team lead

Role

Node-RED development ·
Presentation · Distribution

From input to result: motion-recognition flow



01

Camera input · landmark extraction

- Separated webcam and SmartThings camera inputs with pose-detect-webcam and pose-detect-iotcam
- Extracted body and hand landmarks with MediaPipe BlazePose and Hands
- Passed real-time frame results to the next recognition step

02

Registration · discovery · Node-RED integration

- Saved motion data and searched for similarity with pose-register and pose-find
- Configured hand registration and discovery with hand-register and hand-find
- Delivered recognition results to the automation flow through monitor and WebSocket
- Modularized detection, registration, and discovery for composition in Node-RED

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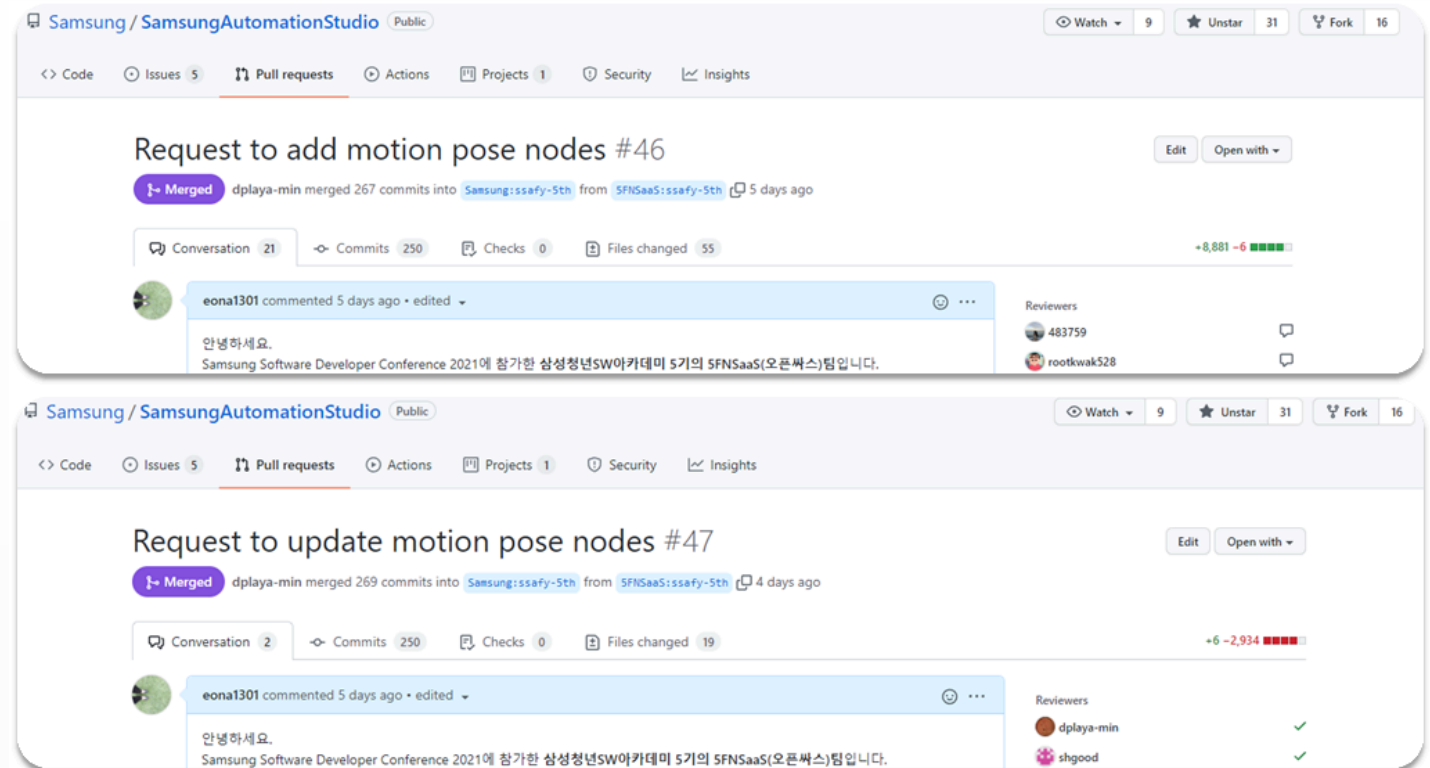
Team

5-person team · Team lead

Role

Node-RED development ·
Presentation · Distribution

Open-source release · contribution · presentation



01

npm · Node-RED Library release

- Published nine motion-recognition nodes as npm package 1.1.3
- Registered the package in the Node-RED Library and supported webcam and SmartThings cameras
- Recorded more than 600 downloads in one week

02

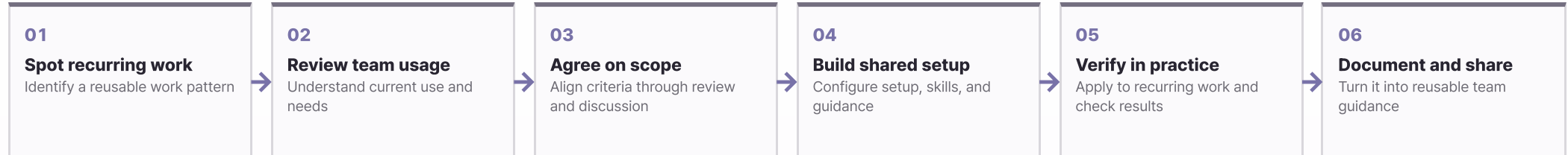
Open-source contribution and presentation

- Merged PRs #46 and #47 into Samsung Automation Studio
- Received 10+ code review comments on public PRs and refined the implementation
- Presented the release, open-source contribution, and IoT use case flow at SSDC

Development Practice and AI

I use Codex and Claude in the development flow, applying changes only after team standards and verification.

AI-assisted decision flow



01 Team Standards

Shared AI Environment

- 01 Structuring shared setup and skills**
Building the base environment and reusable units
- 02 Documenting team usage criteria**
Defining usage principles around context and security

02 Verification and Ownership

Verify Before Applying

- 01 Checking tests, builds, and diffs**
Verifying behavior and impact through execution
- 02 Comparing docs with project code**
Checking fit with the actual version and environment

03 Team Adoption and Scale

Making Recurring Work Shared

- 01 Sharing scope in team discussions**
Defining the shared scope from use cases and feedback
- 02 Preparing Confluence and onboarding materials**
Writing criteria and examples for continued reuse

Key outcome Individual use → Shared team practice

Turned shared setup, skills, and usage criteria into a reusable development environment for recurring work.